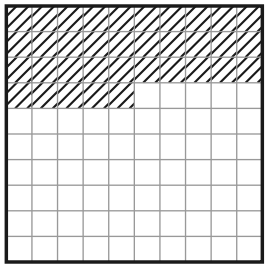
Percents

Percent means “out of 100”.

35% is 35 out of 100



35 of the 100 squares are shaded.

35% = 35/100 = 0.35

Start with 10%: divide by 10

10% of 80 = 8

20% of 80 = 16

5% of 80 = 4

25% of 80 = 20

20% is 2 × 10%. 5% is half of 10%. 25% is 20% + 5% (or a fourth).

Your turn!

30% of 60 = 18

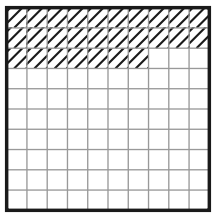
For teachers

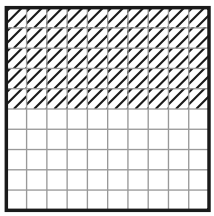
Goal: percent as a rate per 100; percent of a number and the whole from a part (6.RP.3c). Say “Find 10% first, then build the percent you need.”

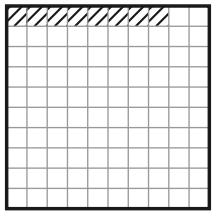
Percents: Practice

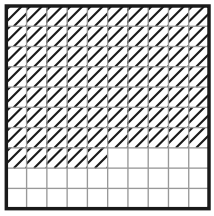
10% first, then build.

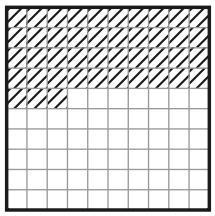
Part 1 Read the grid How many of the 100 squares are shaded?

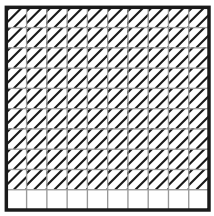
1)  27 % shaded

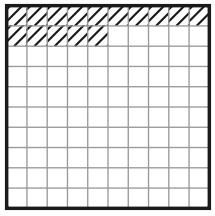
2)  50 % shaded

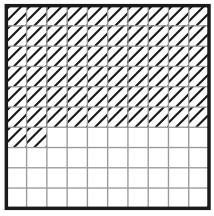
3)  8 % shaded

4)  75 % shaded

5)  43 % shaded

6)  90 % shaded

7)  15 % shaded

8)  62 % shaded

Part 2 10% and 1% Divide by 10, or by 100.

9) 10% of 70 = 7

10) 10% of 250 = 25

11) 10% of 40 = 4

Percents: Practice (continued)

Part 2 10% and 1% (continued)

12) 10% of 860 = 86

13) 10% of 1,200 = 120

14) 10% of 30 = 3

15) 1% of 300 = 3

16) 1% of 2,500 = 25

17) 1% of 700 = 7

18) 1% of 4,100 = 41

19) 1% of 900 = 9

20) 1% of 600 = 6

Part 3 Percent of a number No help this time.

21) 20% of 45 = 9

22) 30% of 70 = 21

23) 50% of 86 = 43

Percents: Practice (continued)

Part 3 Percent of a number (continued)

24) 25% of 60 = 15

25) 5% of 140 = 7

26) 40% of 35 = 14

27) 75% of 80 = 60

28) 15% of 60 = 9

29) 60% of 25 = 15

30) 90% of 30 = 27

31) 35% of 40 = 14

32) 80% of 15 = 12

33) 45% of 20 = 9

34) 70% of 50 = 35

35) 12% of 50 = 6

Percents: Practice (continued)

Part 3 Percent of a number (continued)

36) 25% of 360 = 90

37) 2% of 450 = 9

38) 65% of 20 = 13

39) 55% of 60 = 33

40) 95% of 40 = 38

Part 4 What percent? Make it out of 100.

41) 7 out of 20 = 35 %

42) 3 out of 25 = 12 %

43) 9 out of 10 = 90 %

44) 12 out of 50 = 24 %

45) 1 out of 4 = 25 %

46) 3 out of 5 = 60 %

Percents: Practice (continued)

Part 4 What percent? (continued)

47) 18 out of 25 = 72 %

48) 33 out of 50 = 66 %

49) 7 out of 10 = 70 %

50) 11 out of 20 = 55 %

51) 4 out of 5 = 80 %

52) 21 out of 50 = 42 %

Part 5 Find the whole The part and the percent are given.

53) 10% of 90 = 9

54) 25% of 48 = 12

55) 50% of 74 = 37

56) 20% of 35 = 7

57) 10% of 260 = 26

58) 75% of 40 = 30

Percents: Practice (continued)

Part 5 Find the whole (continued)

59) 5% of 120 = 6

60) 40% of 55 = 22

61) 30% of 70 = 21

62) 25% of 84 = 21

Part 6 More percents of numbers

63) 50% of 300 = 150

64) 60% of 340 = 204

65) 50% of 800 = 400

66) 15% of 460 = 69

67) 5% of 320 = 16

68) 30% of 780 = 234

Percents: Practice (continued)

Part 6 More percents of numbers (continued)

69) 75% of 700 = 525

70) 10% of 180 = 18

71) 50% of 560 = 280

72) 75% of 520 = 390

73) 50% of 260 = 130

74) 30% of 600 = 180

75) 50% of 400 = 200

76) 75% of 160 = 120

77) 50% of 40 = 20

78) 75% of 740 = 555

79) 50% of 600 = 300

80) 40% of 620 = 248

Percents: Practice (continued)

Part 6 More percents of numbers (continued)

81) 25% of 680 = 170

82) 20% of 320 = 64

83) 75% of 560 = 420

84) 30% of 300 = 90

85) 20% of 460 = 92

86) 75% of 300 = 225

Part 7 More finding the whole

87) 50% of 80 = 40

88) 10% of 20 = 2

89) 50% of 52 = 26

90) 25% of 16 = 4

91) 50% of 156 = 78

92) 25% of 100 = 25

Percents: Practice (continued)

Part 7 More finding the whole (continued)

93) 50% of 24 = 12

94) 25% of 60 = 15

95) 50% of 36 = 18

96) 25% of 96 = 24

97) 50% of 136 = 68

98) 50% of 48 = 24

Part 8 Remember? Problems from earlier packets.

99) 120 miles in 4 hours = 30 miles per hour

100) $5 : 2 = 25 : 10$

101) $\frac{1}{5} + \frac{1}{4} = \frac{9}{20}$

102) $0.4 \times 0.4 = 0.16$

103) $2.7\ \text{m} = 270\ \text{cm}$

104) $(18 - 4) \div 7 = 2$

Percents: Practice (continued)

Part 9 Word problems 10% first.

105) A jacket costs $60. It is 25% off. How many dollars do you save?

Number sentence **25% of 60 = 15**

Answer **$15**

Draw or show your work

106) Ben got 18 out of 20 questions right. What percent is that?

Number sentence **18 out of 20 = 90 out of 100**

Answer **90** %

Draw or show your work

107) A class of 30 kids: 40% walk to school. How many kids walk?

Number sentence **40% of 30 = 12**

Answer **12** kids

Draw or show your work

108) Mia has read 20% of her book. That is 45 pages. How many pages are in the book?

Number sentence **20% of 225 = 45**

Answer **225** pages

Draw or show your work

109) Challenge: A $80 game is 15% off. What is the new price?

Number sentence **15% of 80 = 12, 80 − 12 = 68**

Answer **$68**

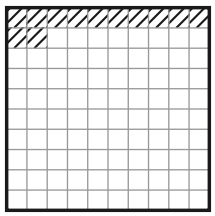
Draw or show your work

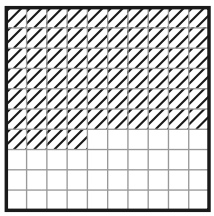
Check-In: Percents

Show what you know. No help this time.

Mastery: 8 of 9 (80%)

Part 1 Read the grid

1)  12 % shaded

2)  64 % shaded

Part 2 Percent of a number

3) 30% of 90 = 27

4) 5% of 160 = 8

5) 40% of 65 = 26

Part 3 What percent?

6) 13 out of 20 = 65 %

7) 9 out of 25 = 36 %

Part 4 Find the whole

8) 50% of 92 = 46

9) 25% of 56 = 14